

Exp No: 24

MODELING CLOCK-DRIVEN CONCURRENT PROCESSES USING SC_THREAD AND WAIT() IN SYSTEMC

Program:

```
#include <systemc.h>

SC_MODULE(ClockExample) {
    sc_in<bool> clk;

    void process() {
        while (true) {
            wait();

            std::cout << "Time: " << sc_time_stamp()
            << " - Clock Triggered" << std::endl;
        }
    }

    SC_CTOR(ClockExample) {
        SC_THREAD(process);
        sensitive << clk.pos();
    }
};

int sc_main(int argc, char* argv[]) {
    sc_clock clock("clock", 1, SC_NS);
    ClockExample module("module");
    module.clk(clock);
    sc_start(5, SC_NS);
    return 0; }
```

Output

Compile done. Starting run...

```
SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
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ALL RIGHTS RESERVED
Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered
```

Done